Botao Ye

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Education

2022.09 - 2026.06 (Expected)

Kyungpook National University, School of IT, Global Software Convergence Major (South Korea)

- One of South Korea's National Flagship Universities, ranked No. 91 in Asia and No. 1 among Korean national universities in the 2025 THE Asia Rankings, and =519 in the 2026 QS World University Rankings, where I received solid training in software engineering, AI, and research-oriented systems development.
- Average Score: **90.4/100**
- Selected coursework: Artificial Intelligence, Fundamentals of Data Science, Data Structures, Computer Architecture, Introduction to Neural Networks, Software Design, Open Source Programming

Selected Research Highlights

- **Independent problem formulation and ownership:** All three research projects were independently formulated by me. For **CuraView** and **Decathlon VOC Analyzer**, aside from necessary collaboration, I carried out most of the core work from problem definition and technical route design to system implementation, experiments, and paper preparation. **Craft-in-the-Loop** began as a problem I independently proposed and a first version I developed, and was later expanded through group collaboration into formal research.
- **Trustworthy medical AI with GraphRAG:** Designed CuraView, a multi-agent medical hallucination detection framework for discharge summaries. On 1,103 test sentences, the safety-critical E4 detector achieved a **50% relative improvement** and performed **15% better than mainstream closed-source models**. The manuscript is **under review at KBS**, and the preprint is available on **arXiv**.
- **Long-form multi-agent generation:** Built Craft-in-the-Loop, a Planner/Writer framework for structured long-form text generation with shared memory and narrative event trees. The paper was **accepted by ADHO**.
- **Evidence-driven multimodal systems:** Developed Decathlon VOC Analyzer, a multimodal RAG and claim-level attribution system for product analysis. The project passes **166 automated tests**, and the paper was **accepted by KIIT**.
- **Current research direction:** I am particularly interested in **trustworthy agent memory**, especially how latent internal state, explicit memory cards, and auditable behavior traces can be aligned in long-running AI systems.

Selected Publications and Systems

- **CuraView: Medical Hallucination Detection and GraphRAG-Based Evidence Verification for Discharge Summaries**
  - Status: **arXiv preprint**; manuscript **under review at KBS**
  - System role: Multi-agent medical verification system that combines patient-level EHR evidence graphs, GraphRAG retrieval, sentence-level claim checking, and structured evidence grading.
  - Links: arXiv: arxiv.org/abs/2605.03476 | GitHub: github.com/severin-ye/CuraView
- **Craft-in-the-Loop / Narrative-Causal-Graph: Narrative Schema-Based Multi-Agent Generation for Long-Form Text**
  - Status: **Accepted by Alliance of Digital Humanities Organizations (ADHO)**
  - System role: Shared-memory multi-agent generation system that maintains long-range plot consistency through chapter planning, event trees, and structured narrative states.
  - Link: GitHub: github.com/Xynovitch/Narrative-Causal-Graph
- **Decathlon VOC Analyzer: Evidence-Driven Multimodal VOC Analysis with Claim-Level Attribution**
  - Status: **Accepted by Korean Institute of Information Technology (KIIT)**
  - System role: Multimodal evidence attribution system with artifact-centric experiment management, replay files, run manifests, and claim-level citation for reproducible analysis.
  - Link: GitHub: github.com/severin-ye/Decathlon_VOC_Analyzer

Research Experience

CuraView: Medical Hallucination Detection and GraphRAG-Based Evidence Verification

2025.04 - 2026.01

- Designed a multi-agent framework for sentence-level hallucination detection in medical discharge summaries, focusing on factual consistency risks in high-stakes clinical text generation.
- Built a patient-level GraphRAG pipeline over diagnoses, medications, laboratory results, timestamps, numerical records, negation relations, and contradictory evidence.
- Developed a seven-category hallucination taxonomy and an E1-E4 evidence grading scheme to support interpretable review and downstream training.
- Combined Qwen3-14B fine-tuning, structured JSON output, and Pydantic validation to improve robustness in batch inference.

Craft-in-the-Loop: Narrative Schema-Based Multi-Agent Generation for Long-Form Text

2025.06 - 2026.02

- Proposed a Planner/Writer architecture that converts narrative schema into chapter plans, stasis/process statements, and hierarchical event trees.
- Built a reference corpus of 100 bestselling novels and generated 100 long-form texts exceeding 100,000 tokens for comparative evaluation.
- Designed a six-dimensional evaluation framework covering originality, diversity, reader engagement, plot consistency, characters and settings, and spatiotemporal clarity.
- Showed that structured mid-level narrative control reduces long-form drift and homogenization compared with vanilla generation baselines.

Decathlon VOC Analyzer: Evidence-Driven Multimodal VOC Analysis System

2025.07 - 2026.05

- Developed a multimodal RAG pipeline for aligning reviews, product text, product images, and local visual regions with claim-level evidence attribution.
- Implemented aspect-level review modeling, question planning, text/image retrieval, reranking, structured report generation, and artifact-based replay.
- Built reproducibility support through evidence packs, retrieval logs, report claims, sidecars, and manifests.
- Established validation coverage across normalization, modeling, retrieval, report generation, and workflow entry points, reaching 166 automated tests.

Entrepreneurial And Project Management Experience

AI Review Assistant Development, Hainan Zhiyi Health Management Co., Ltd.

2023.10 - 2024.05

- Drove the project in an entrepreneurial setting, covering client communication, requirement definition, prototype design, development coordination, and deployment of an AI-assisted document review tool.
- Managed product goals and technical execution from a project-manager perspective, while improving medical document verification and correction through model integration and LoRA-based optimization.

Recommendation Algorithm Design And Development, Chengdu Kamang Network Technology Co., Ltd.

2025.01 - 2025.02

- Worked in a project-management role for a game recommendation initiative, coordinating requirement analysis, recommendation logic design, implementation, and iterative refinement.
- Built multidimensional matching features for retrieval-enhanced and interpretable recommendation, and applied LoRA-based optimization on Qwen-32B to improve output quality.

Technical Skills

- **Programming:** Python, Flask, C, Django, Vue
- **AI / ML:** PyTorch, Transformer, RAG/GraphRAG, LoRA/QLoRA, structured output constraints, preference optimization
- **Retrieval / Multimodal:** Qdrant, vector indexing, CLIP and vision-language embeddings, multi-stage retrieval, reranking, claim-level attribution
- **Engineering:** Git, GitHub, Linux, Docker, LangChain, MS-SWIFT, Pydantic, automated testing, reproducible experiment management
- **Languages:** Chinese (native), Korean (TOPIK Level 4), English (TOEFL planned; mock score 80+)

Future Research Interests

- **Trustworthy agent memory:** explicit memory cards, auditable behavior traces, and long-term preference or policy retention in AI agents.
- **Evidence-driven medical AI:** factual verification, clinical evidence workflows, structured outputs, and interpretable high-risk AI systems.
- **Multi-agent planning and long-context generation:** shared memory, event-structured planning, and controllable generation in long-running tasks.

Awards

- **Kyungpook National University Excellent International Student Scholarship**: six consecutive semesters, including three first-class scholarships (top 8%).
- **EY ESG Regional Competition - Third Prize (2023)**.